Department Of Electrical Engineering

Summer Internship Report

Name of Intern – Shivam Maurya

Institute – IIT Bhilai

Mentor – Dr. Olive Ray

Duration of internship – 4 Weeks (14 May 2026 to 10 June 2026)

Design, Simulation, and Experimental Characterization of a SiC MOSFET Gate Driver System

Time Line of Tasks –

- 14 May: Joining
- 14 May – 16 May: Altium Installation and BaSICs
- 16 May – 19 May: Made the schematic of STGAP2SICSNC isolated 4 A single gate driver using the Datasheet.
- 19May- 25May: Learned the PCB baSICs and then Designed a two-layer PCB for the same schematic in Altium. Also, made the BJT part of circuit of Murata power supply for analysis in LT spice. Concluded its working as an emitter follower.
- 25May – 26May: learned about Gerber files and how to export the PCB design for fabrication.
- 26May – 28May: Simulated the Gate driver circuit on LT spice and got the switching node waveform as expected.
- 28May – 31May: Learned about code composer studio and the microcontroller TMS320F28379D's programming logic.
- 1June – 8June: Made the code for continuous pulse signal with deadband and used it first for checking the gate driver board, then performed the pulse test and captured the switching waveform and calculated the switching loss which were verified from the oscilloscope readings.
- 8June – 10June: Applied complementary signals at both the switches to capture the complementary switching waveform and also developed the code of double pulse.

Introduction

This report presents the work carried out during a four-week summer internship at the Department of Electrical Engineering, IIT Bhubaneswar, from 14 May 2026 to 10 June 2026, under the guidance of Dr. Olive Ray. The internship focused on the design, simulation, and experimental characterization of power electronics circuits, with a specific emphasis on Silicon Carbide (SiC) MOSFET-based gate driver systems.

Wide Bandgap (WBG) semiconductor devices such as SiC MOSFETs have emerged as a preferred choice in modern power conversion systems owing to their superior switching speed, higher breakdown voltage, and lower conduction losses compared to conventional silicon devices. However, their fast-switching characteristics also introduce challenges such as parasitic ringing and precise gate drive requirements, making their experimental characterization an important area of study.

The internship covered a broad range of practical tasks: schematic design and PCB layout of an isolated gate driver circuit using the STGAP2SICSNC gate driver IC in Altium Designer, circuit simulation in LTspice, embedded firmware development for the TMS320F28379D C2000 microcontroller using Code Composer Studio (CCS), and hardware testing including continuous pulse generation, deadband signal verification, and the Double Pulse Test (DPT) for switching loss characterization of SCT35N65 SIC MOSFETs. Switching energies were measured using the Yokogawa DLM3054 oscilloscope.

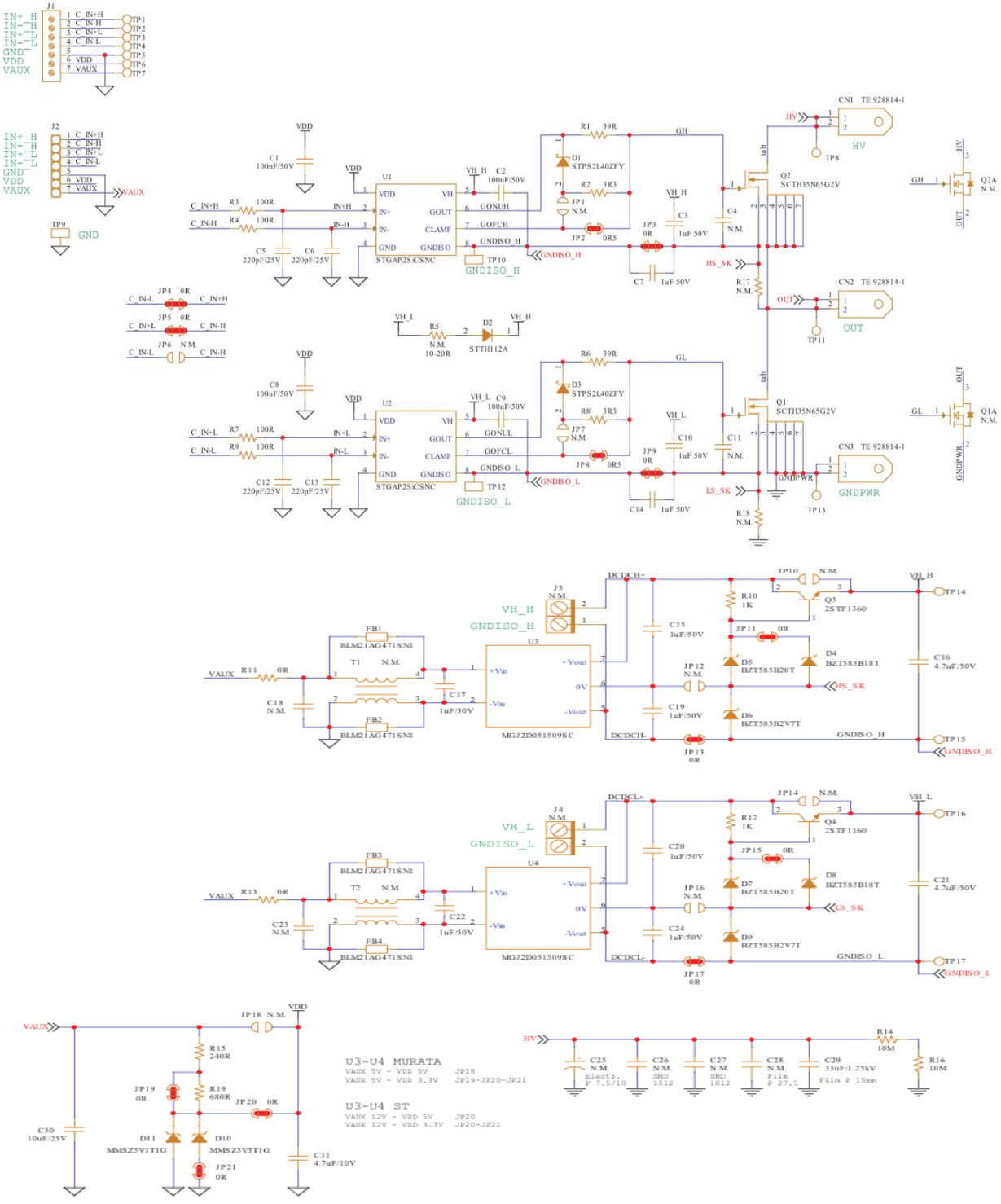
This report documents the timeline of tasks, the technical methodology followed, key experimental results, and the challenges encountered along with how they were resolved.

Brief Summary of Each Task

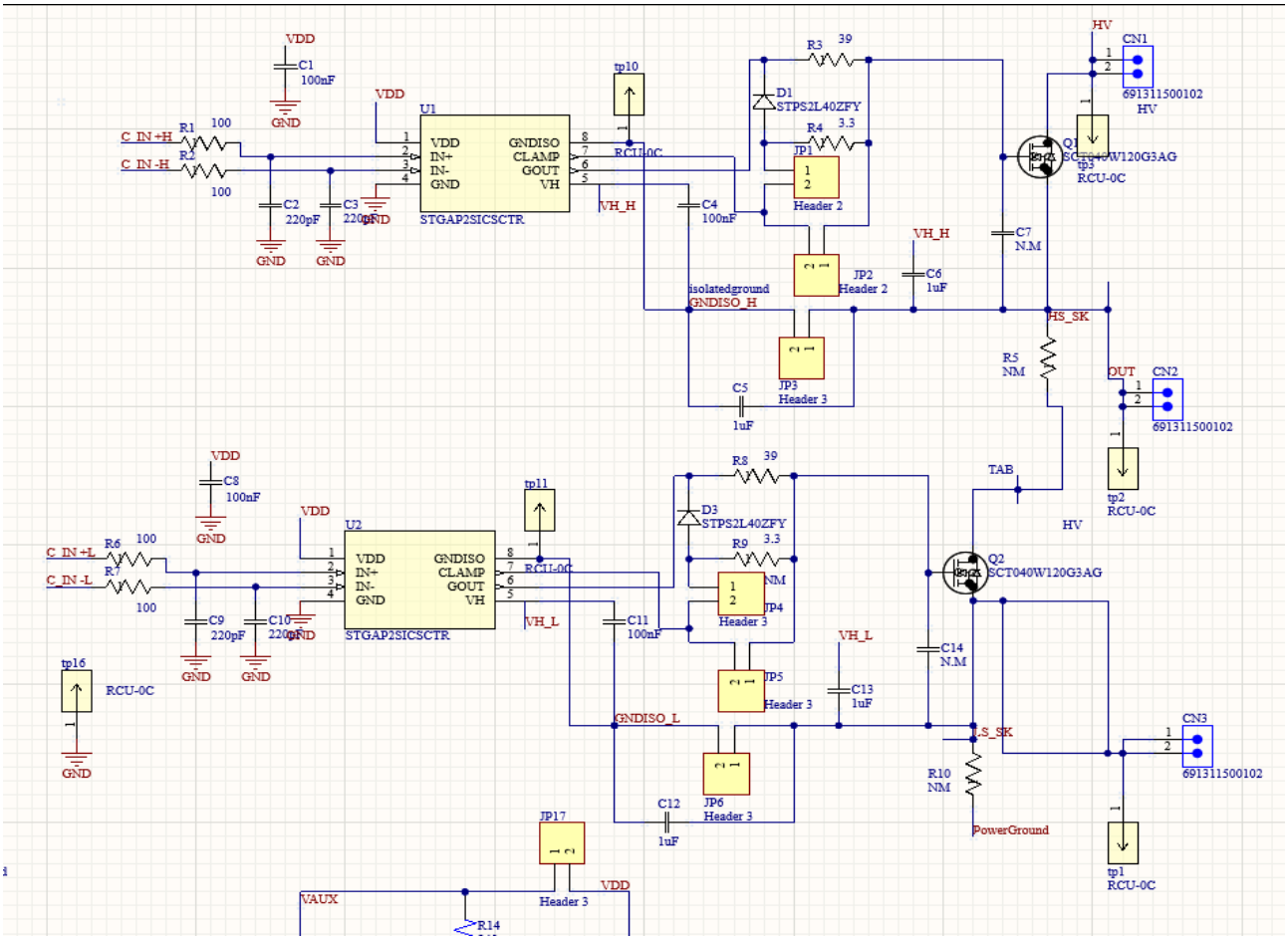
1 - Altium Designer – Schematic & PCB Basics

During the initial phase, I learned Altium Designer, a professional PCB design software widely used in the industry. I learned how to source and add components from online libraries such as Ultra Librarian and Element14, ensuring the correct symbols and footprints were imported. I also understood the importance of verifying component footprints against actual fabrication constraints to avoid design errors during manufacturing. Additionally, I gained familiarity with various passive and active components used in PCB design, including connectors, headers, test points, jumpers, and other standard circuit elements.

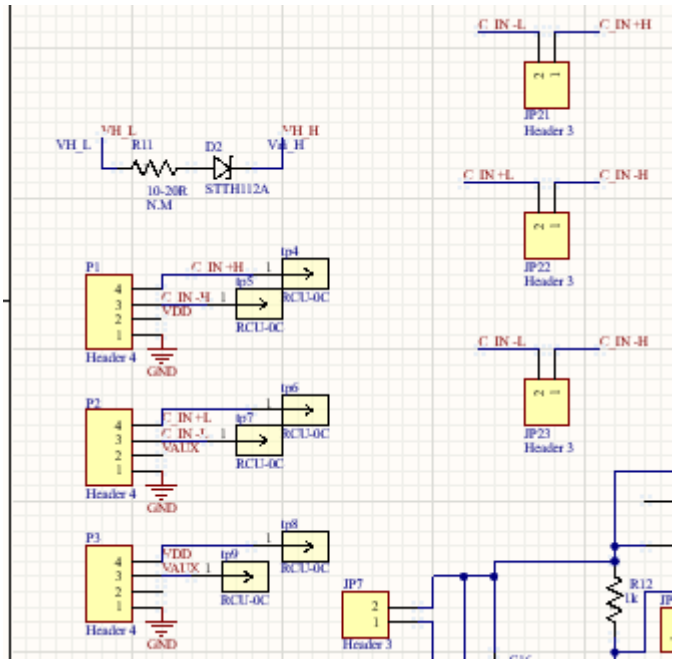
The following figure is from the data sheet of the gate driver



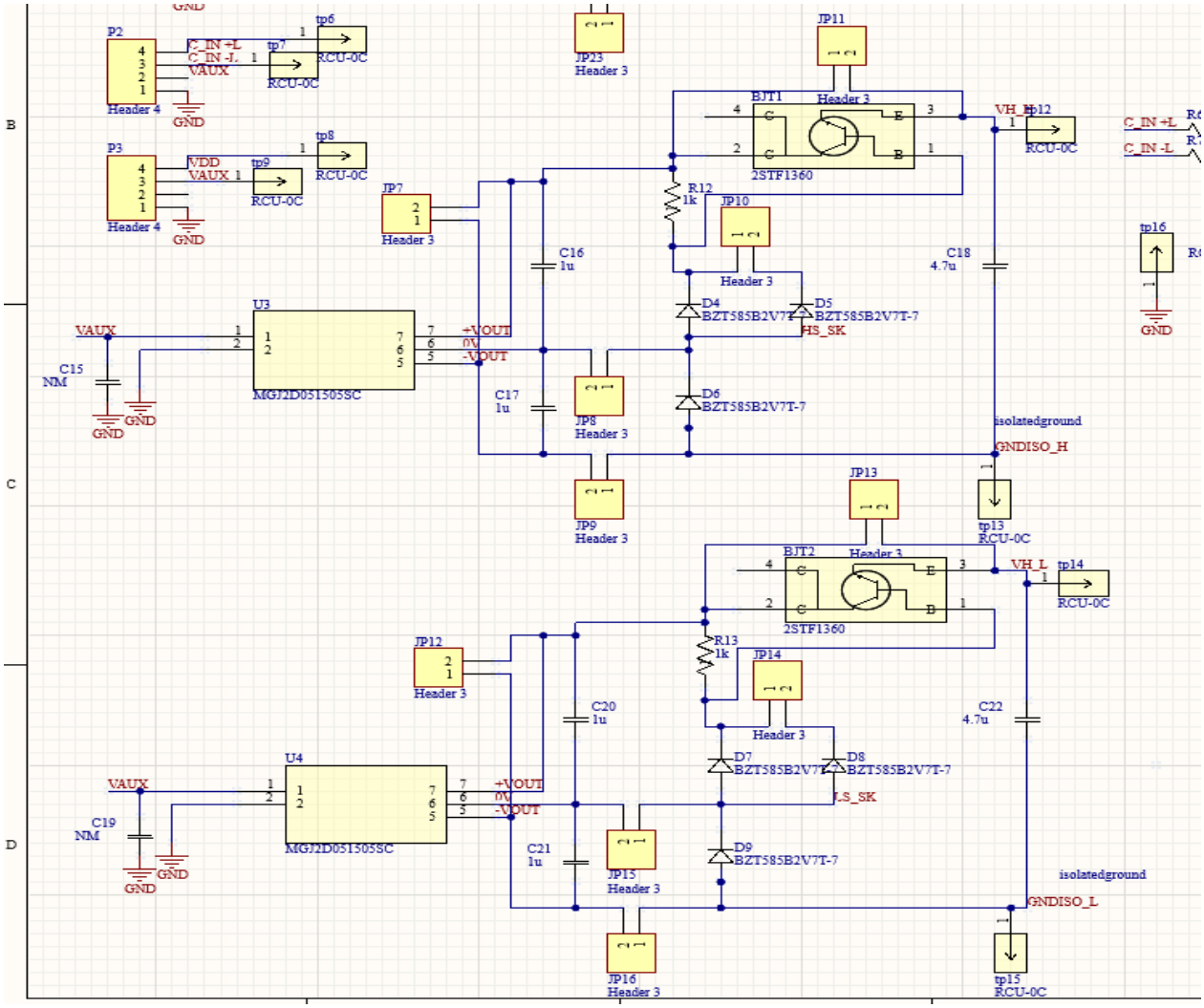
I used this as a reference for my Altium schematic and made this schematic in Altium.



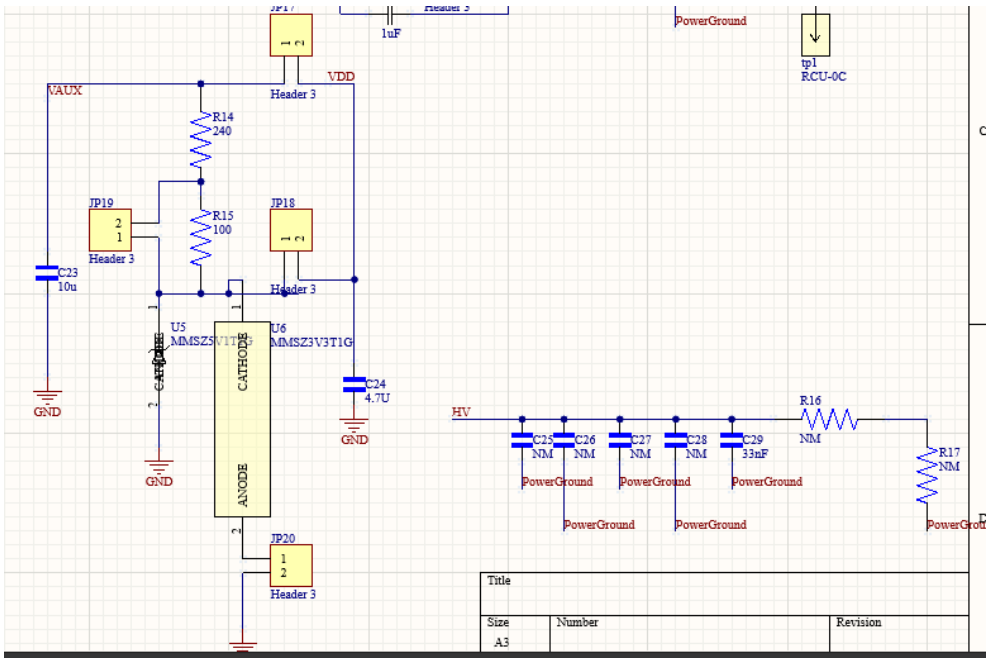
This figure shows the switch and gate driver part of the overall circuits. The Q1 and Q2 MOSFETs are our switches. U1 and U2 are the gate driver IC.



This figure shows the connectors used to give input [Cin+h, Cin-h, Cin+l, Cin-l] and biasing [Vdd, Vaux] to the board also the test points across which we can connect the probe to analyse the working.



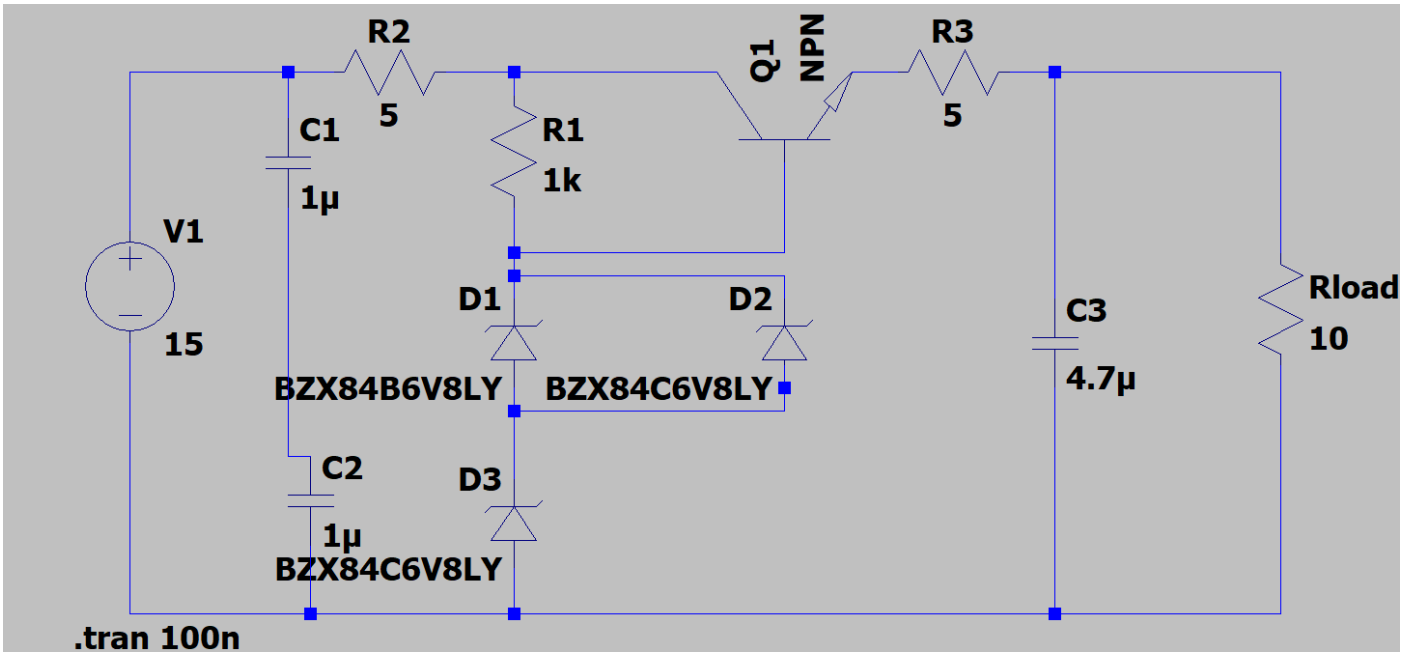
This figure shows the MURATA IC that is used to generate the HS_SK, LS_SK signals, these are the signals connected to cn1, cn2, cn3 across which we take the output.

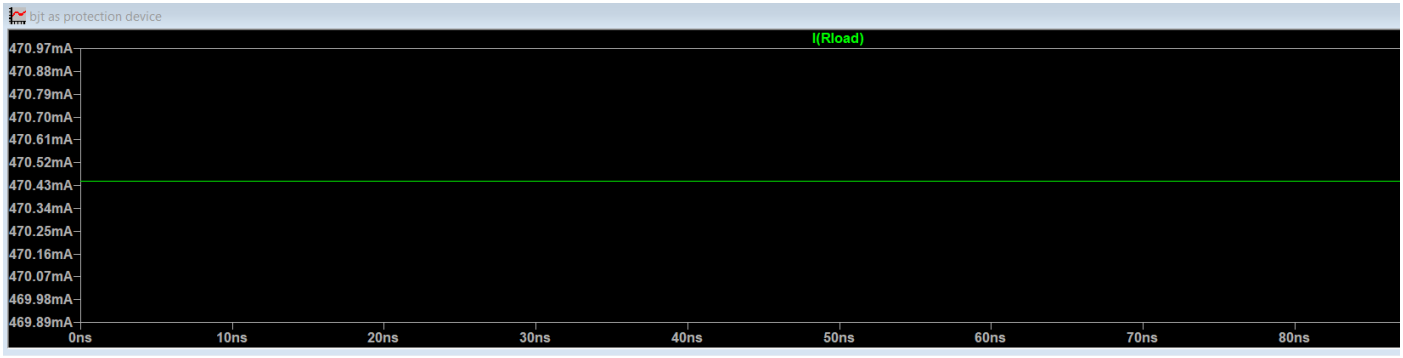


This figure shows a voltage divider between Vaux and Vdd and a low pass filter for Hv to reduce the noise.

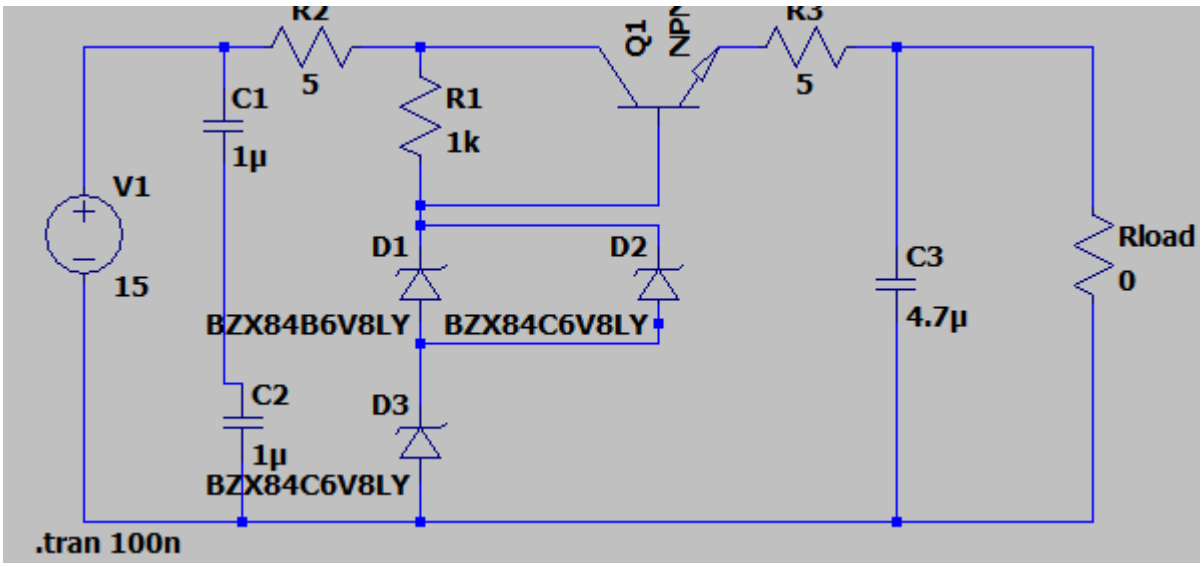
After completing this schematic, I made the BJT circuit (shown in the Murata figure) in LT spice to judge what was the purpose of a BJT there.

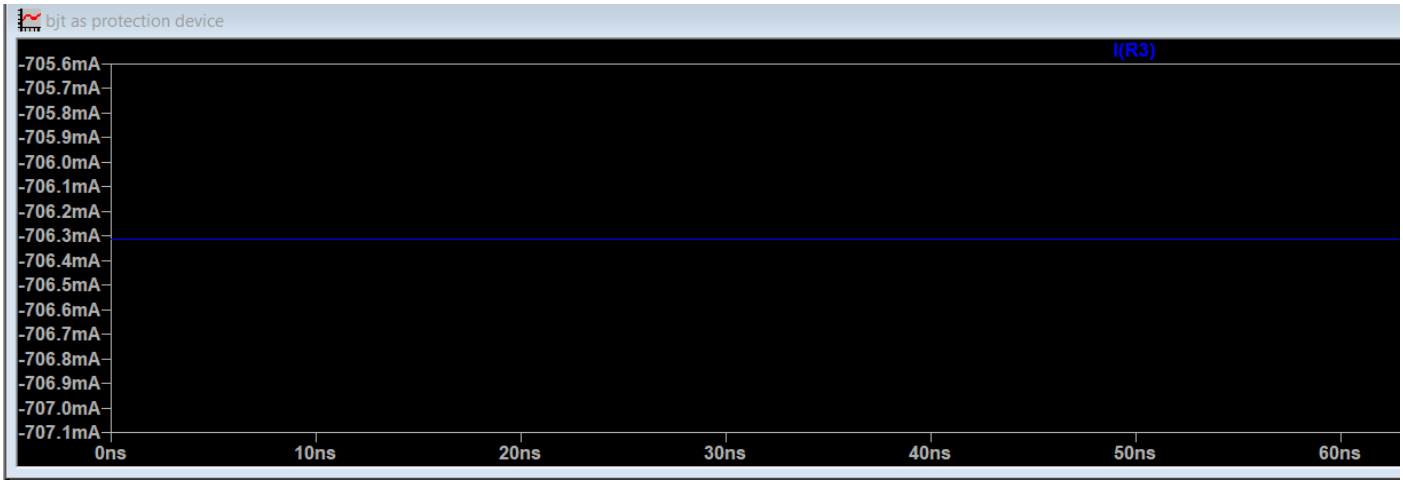
Schematic prepared in LT spice –





When we set the Rload = 0.





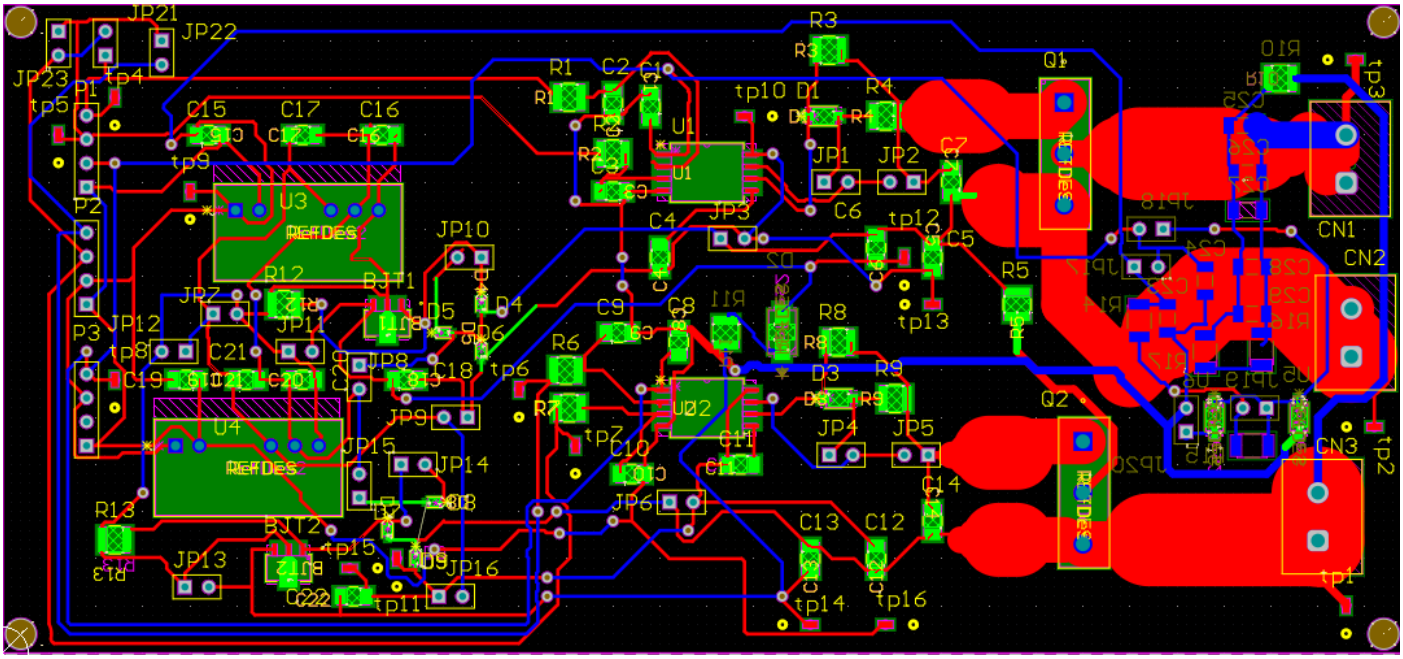
So, the current value remains almost constant when Rload is varied.

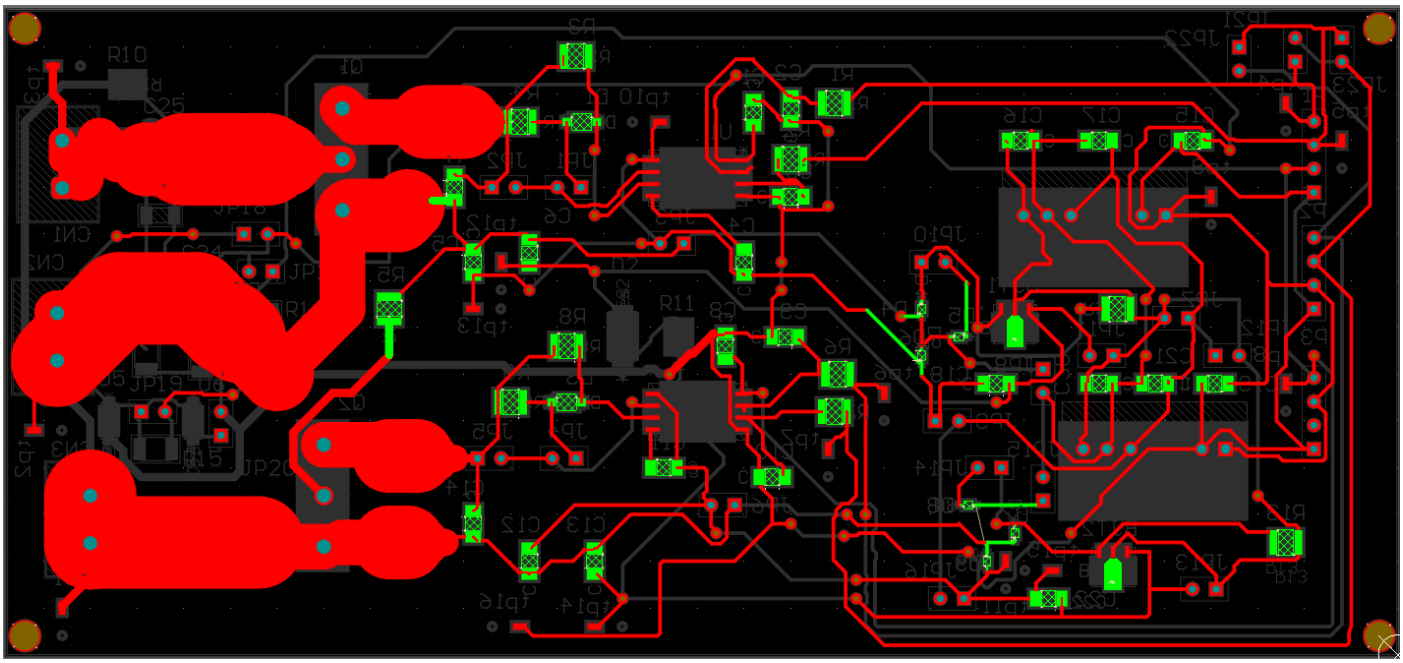
The emitter follower provides current gain and buffering while maintaining approximately unity voltage gain.

After completing the schematic and analysis, I proceeded to design the PCB layout for the gate driver circuit in Altium Designer. This phase gave me hands-on experience with the practical considerations and design rules that govern a good PCB layout. The key concepts I learned and applied are as follows:

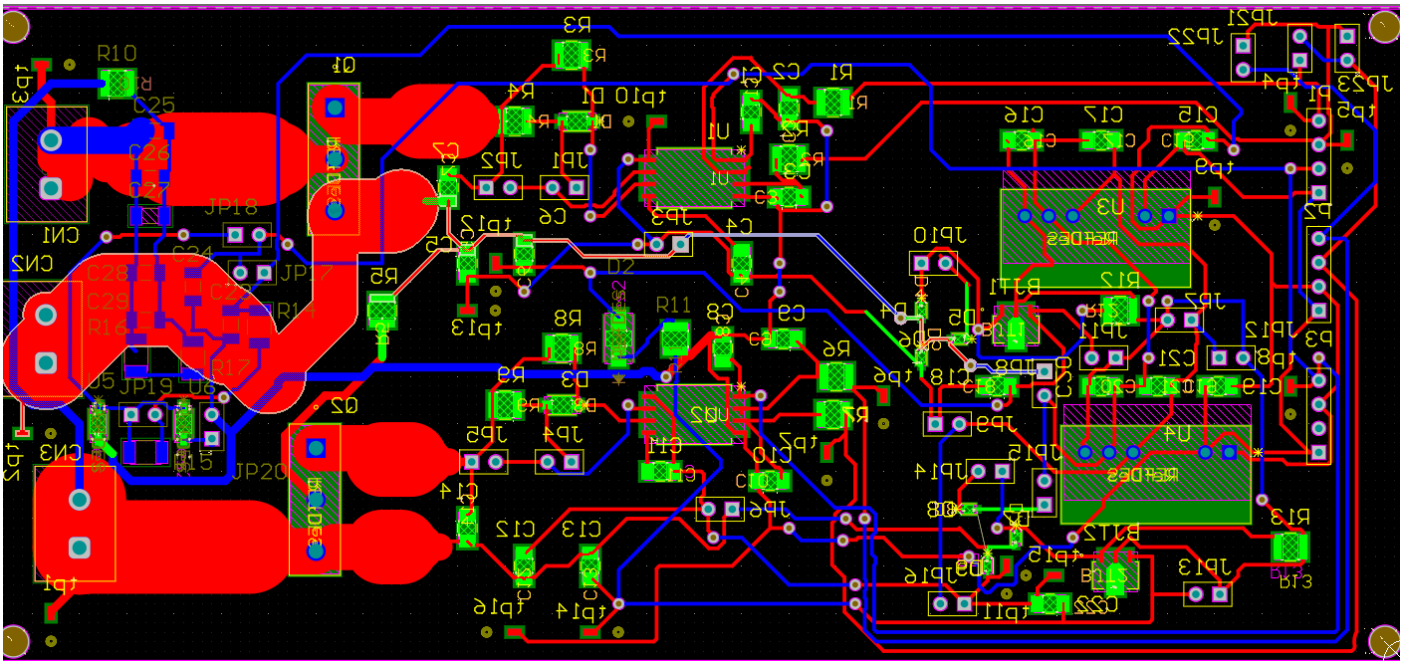
- **Component Placement:** Kept input resistors and capacitors as close as possible to the gate driver IC to minimize parasitic inductance, which is critical in high-speed switching circuits to avoid unwanted voltage spikes and signal distortion.
- **Edge Placement of Connectors:** Positioned all output connectors along the edges of the board to ensure better airflow and thermal management, as well as easier access during testing and integration.
- **Symmetrical Layout:** Maintained board symmetry wherever possible to ensure uniform current distribution, reduce electromagnetic interference (EMI), and simplify the fabrication and assembly process.
- **Input Header Accessibility:** Placed all input headers (J1, J2) along the board edges so that they are easily accessible on the phySICal board during testing, debugging, and wiring.
- **Two-Layer PCB:** Designed the board as a two-layer PCB, utilizing the top layer for signal routing and the bottom layer for voltage divider, which helps in noise reduction and signal integrity.
- **Design Rule Check (DRC):** Ran DRC checks in Altium to verify that all clearances, trace widths, and via sizes met the fabrication constraints before finalizing the design.
- **Gerber File Export:** Learned how to generate and export Gerber files from Altium, which are the standard files sent to PCB manufacturers for fabrication.
- Importance of cross section area of wires according to the amount of current they are carrying.

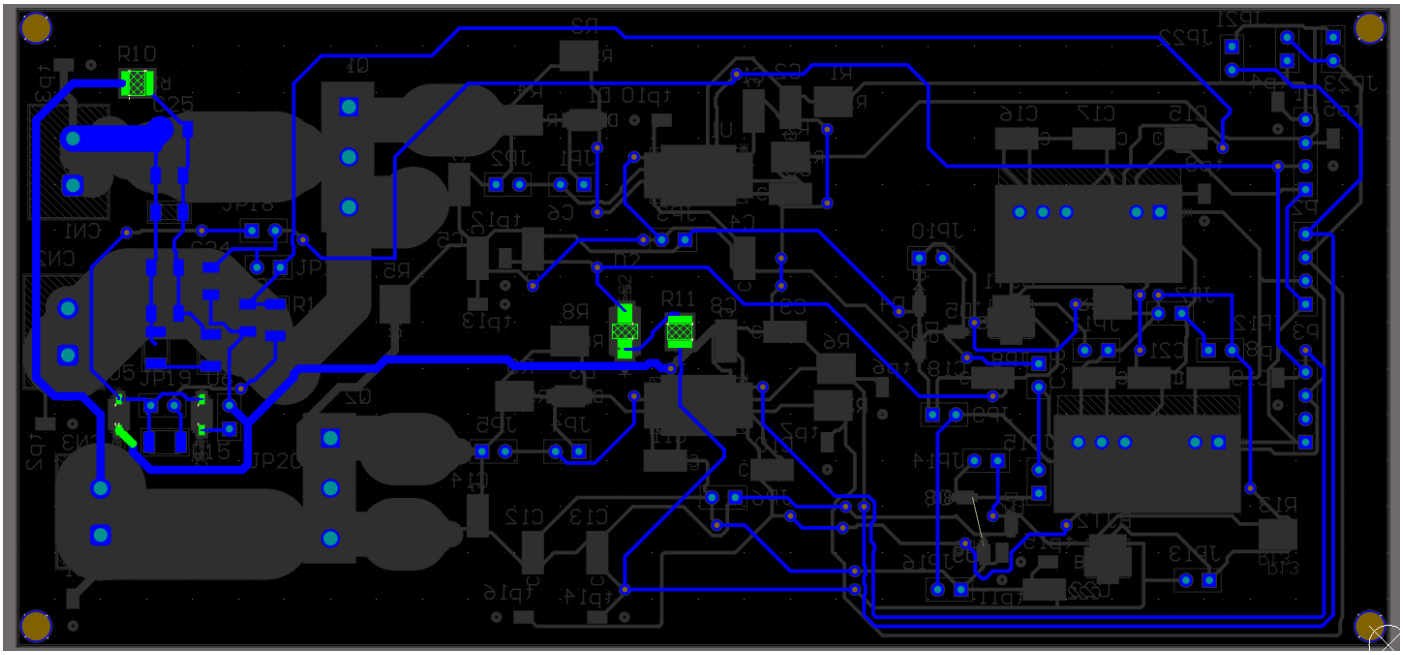
This is the top view of my pcb

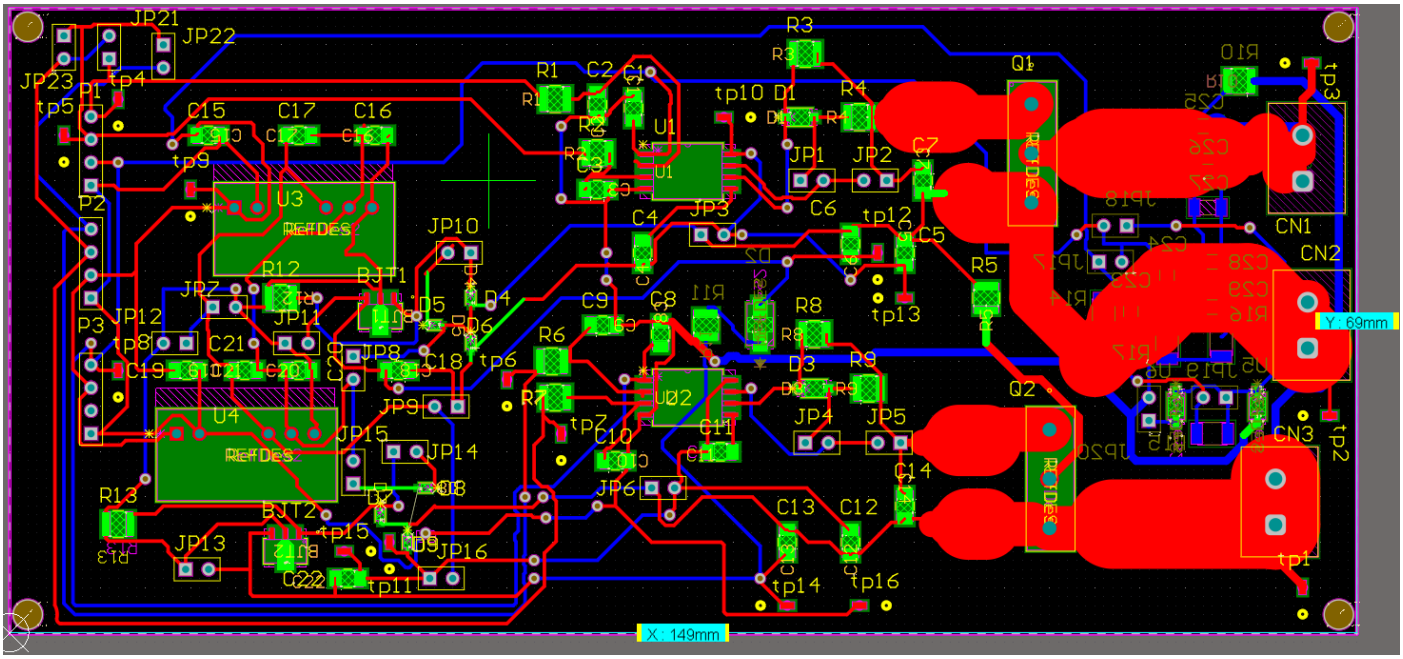




This is the bottom view of my PCB.



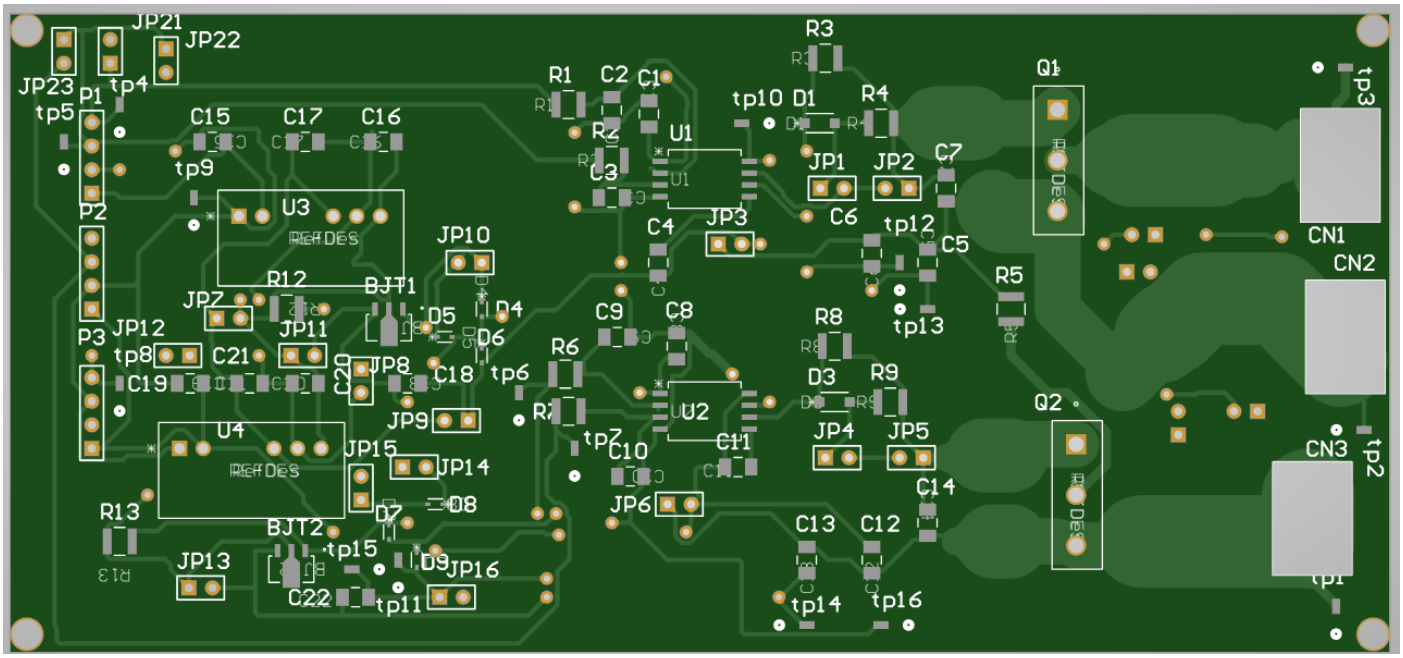




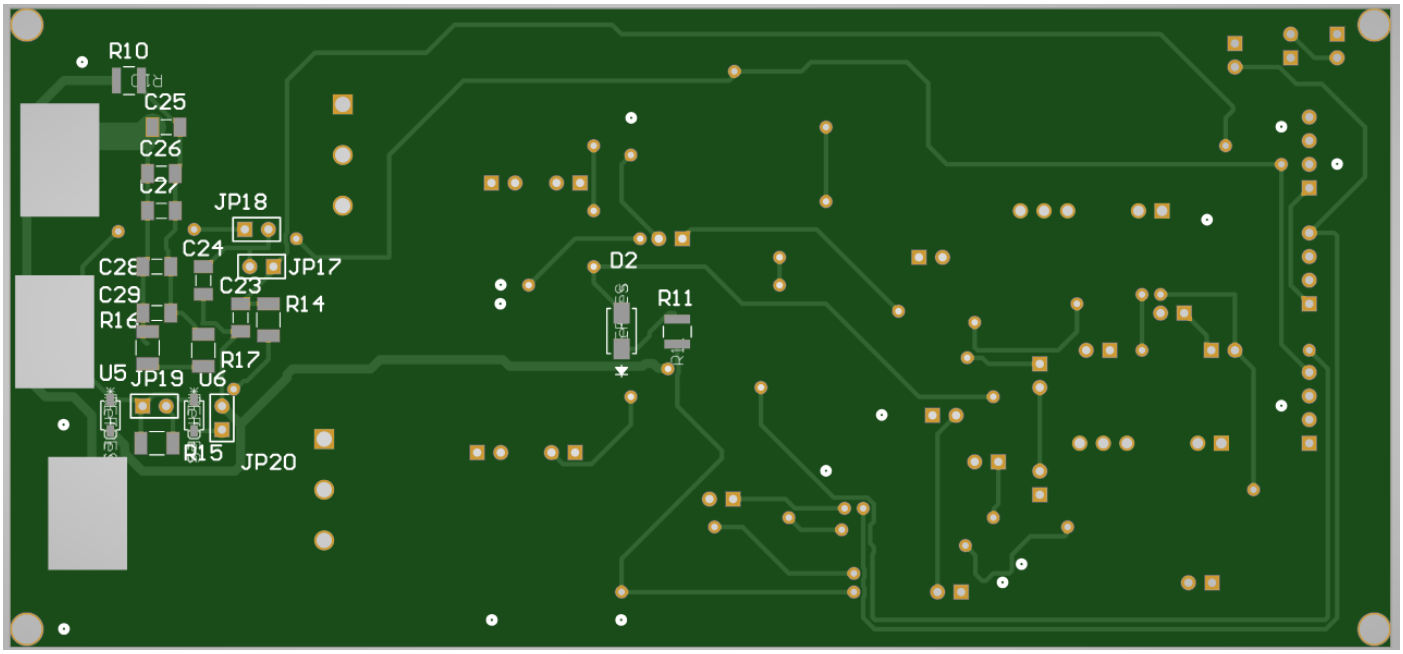
Size of My PCB = 149 * 66 mm

After this I verified this on a website called PCBway.com that shows how your pcb will look like after fabrication using the Gerber files.

Top view:

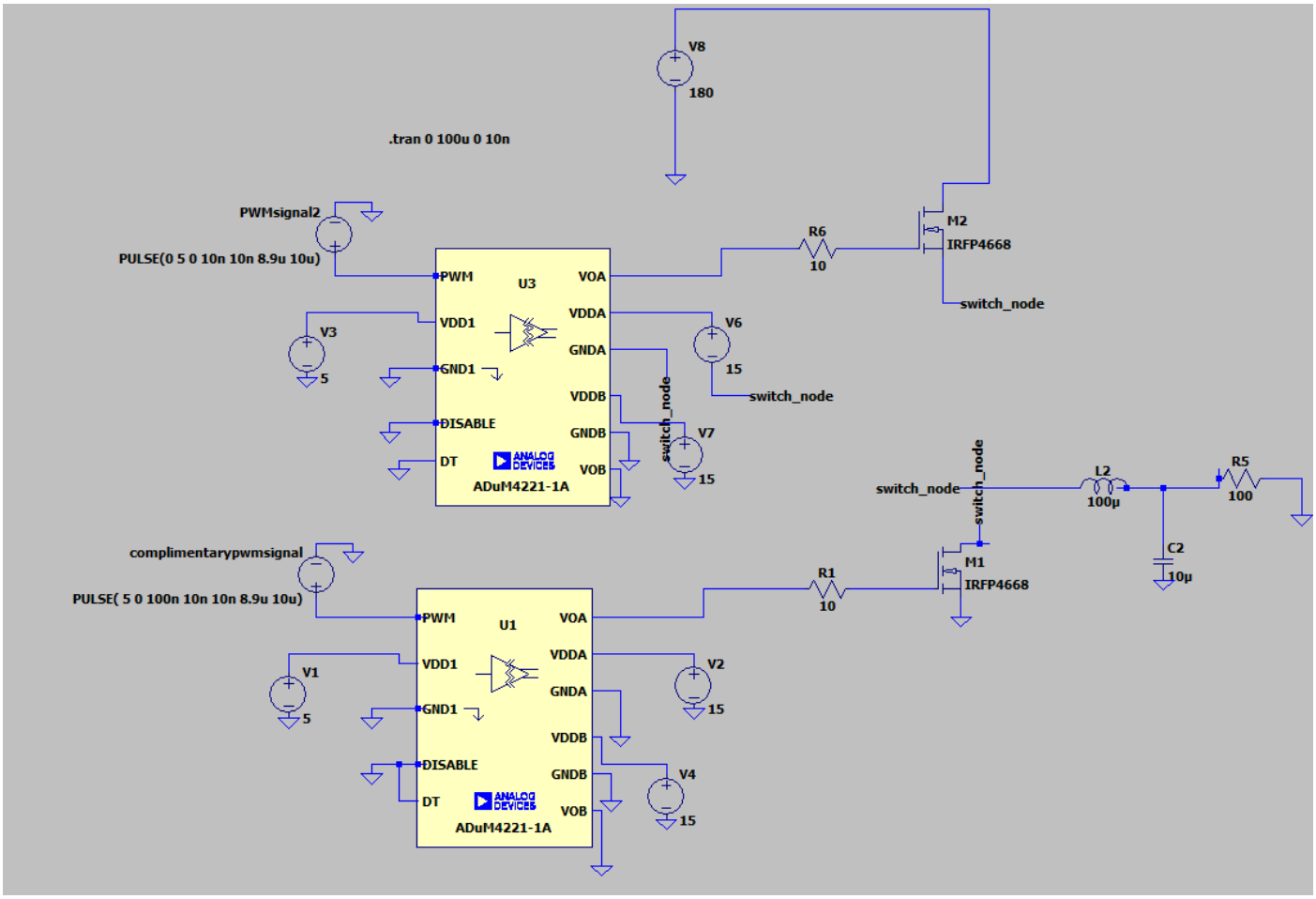


Bottom View:



After completing this I tried to analyse this gate driver circuit on LTSpice. But I ran into a problem.

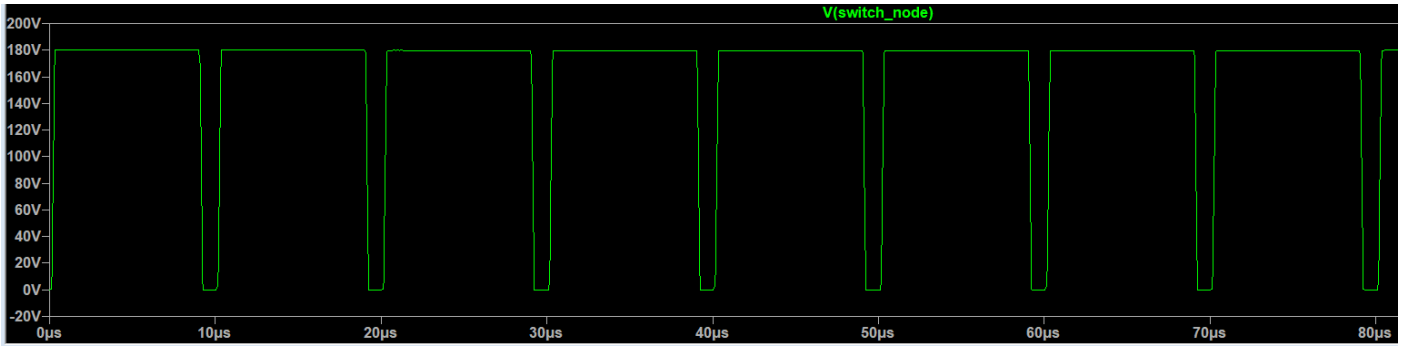
The model of MOSFET (SCTH35N65G2V) and gate driver (STGAP2SICSNC) that we were using are not available in spice library. So, I used a similar silicon MOSFET and a gate driver of similar rating to simulate the working.



This is the schematic I used for analysis.

Mos used - IRP4668

Gate Driver – AduM4221-1A

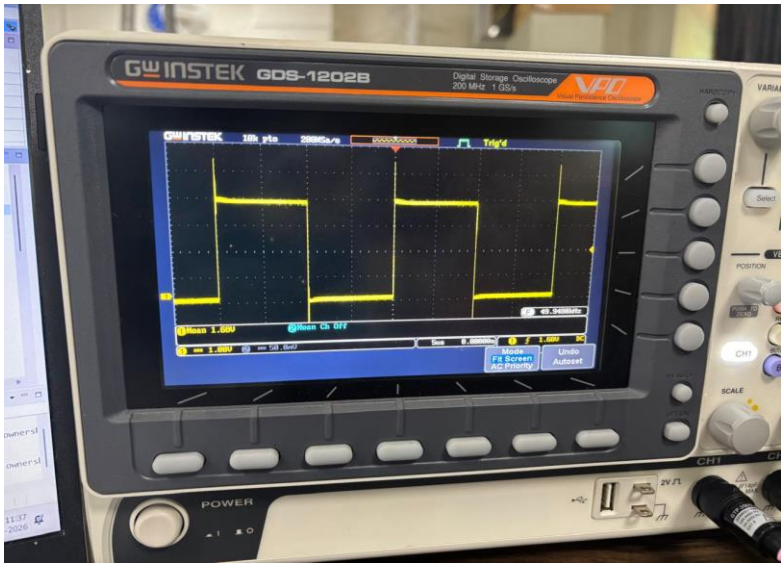


A Clean pulse that oscillates between 180 V and 0V was obtained at switching node.

This can be used to turn the switch on and off at required frequency easily.

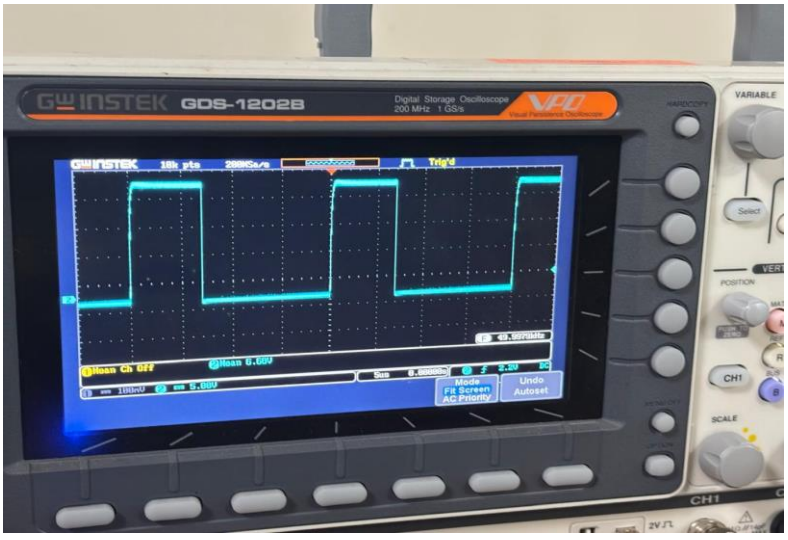
After this I tried to perform the double pulse test on gate driver board STGAP2SICSNC manufactured by ST Microelectronics.

Firstly, I Verified if the board is working properly. For that I applied a pulse of 50 kHz at one of switches keeping the other open.



f = 50Khz

V = 1.68 Volts (input to gate)



F = 50kHz

V = 6.66V (amplified by the gate driver)

I tested like this on various frequencies like 100kHz, 120kHz, etc. and got similar amplification.

Code used for this

```
#include "F28x_Project.h"

void main(void)
{

    InitSysCtrl();


    EALLOW;
    ClkCfgRegs.PERCLKDIVSEL.bit.EPWMCLKDIV = 1;
    EDIS;

    InitGpio();

    EALLOW;

    GpioCtrlRegs.GPAMUX1.bit.GPIO0 = 1;
    GpioCtrlRegs.GPADIR.bit.GPIO0  = 1;
    GpioCtrlRegs.GPAPUD.bit.GPIO0  = 0;


    GpioCtrlRegs.GPAMUX1.bit.GPIO1 = 1;
    GpioCtrlRegs.GPADIR.bit.GPIO1  = 1;
    GpioCtrlRegs.GPAPUD.bit.GPIO1  = 0;
    EDIS;


    DINT;
    InitPieCtrl();
    IER = 0x0000;
    IFR = 0x0000;
    InitPieVectTable();


    EALLOW;
    CpuSysRegs.PCLKCR2.bit.EPWM1     = 1;
    CpuSysRegs.PCLKCR0.bit.TBCLKSYNC = 0;
    EDIS;

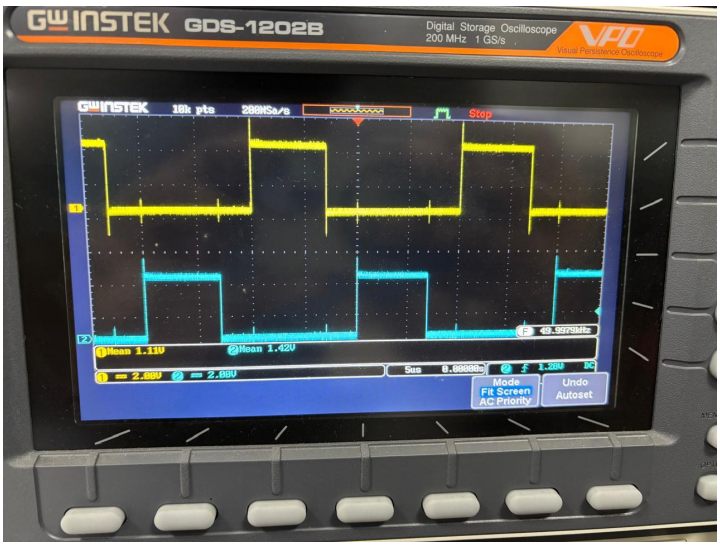

    EPwm1Regs.TBCTL.bit.CTRMODE    = TB_COUNT_UP;
    EPwm1Regs.TBCTL.bit.HSPCLKDIV = TB_DIV1;
    EPwm1Regs.TBCTL.bit.CLKDIV     = TB_DIV1;
    EPwm1Regs.TBCTL.bit.PHSEN      = TB_DISABLE;
    EPwm1Regs.TBCTL.bit.PRDLD      = TB_SHADOW;
    EPwm1Regs.TBCTR                = 0;
    EPwm1Regs.TBPRD                = 999;
```

```
48
49
50      EPwm1Regs.CMPCTL.bit.SHDWAMODE = CC_SHADOW;
51      EPwm1Regs.CMPCTL.bit.LOADAMODE = CC_CTR_ZERO;
52      EPwm1Regs.CMPA.bit.CMPA        = 500;
53
54
55      EPwm1Regs.AQCTLA.bit.ZRO = AQ_SET;
56      EPwm1Regs.AQCTLA.bit.CAU = AQ_CLEAR;
57
58
59      EPwm1Regs.AQCTLB.bit.ZRO = AQ_CLEAR;
60      EPwm1Regs.AQCTLB.bit.CBU = AQ_SET;
61
62
63      EPwm1Regs.DBCTL.bit.OUT_MODE = DB_FULL_ENABLE;
64      EPwm1Regs.DBCTL.bit.POLSEL   = DB_ACTV_HIC;
65      EPwm1Regs.DBCTL.bit.IN_MODE  = DBA_ALL;
66
67
68      EPwm1Regs.DBRED.bit.DBRED    = 150;
69
70
71      EPwm1Regs.DBFED.bit.DBFED    = 150;
72
73
74      EALLOW;
75      CpuSysRegs.PCLKCR0.bit.TBCLKSYNC = 1;
76      EDIS;
77
78
79      EINT;
80      ERTM;
81
82
83      while(1);
84  }
```

This is the deadband code that generates two pulses with a deadband between them through appropriate values of phase shift, duty cycle.

Pulse 1 generated at epwm1A – pin 40 of TMS320F28379D

Pulse 2 generated at epwm1B – pin 39 of TMS320F28379D



So, I just used the epwm1a – pin 40 for this verification.

 EPWMCLKDIV = 1 → EPWM clock = **100 MHz / 2 = 50 MHz** (divides by 2)

 TBPRD = 999 → 1000 counts per period

 Period = 1000 × (1/50MHz) = 1000 × 20ns = **20 µs**

 Frequency = 1/20µs = **50 kHz**

After this for generating the 100kHz pulse, set TBPRD = 500 and CMPA = 250

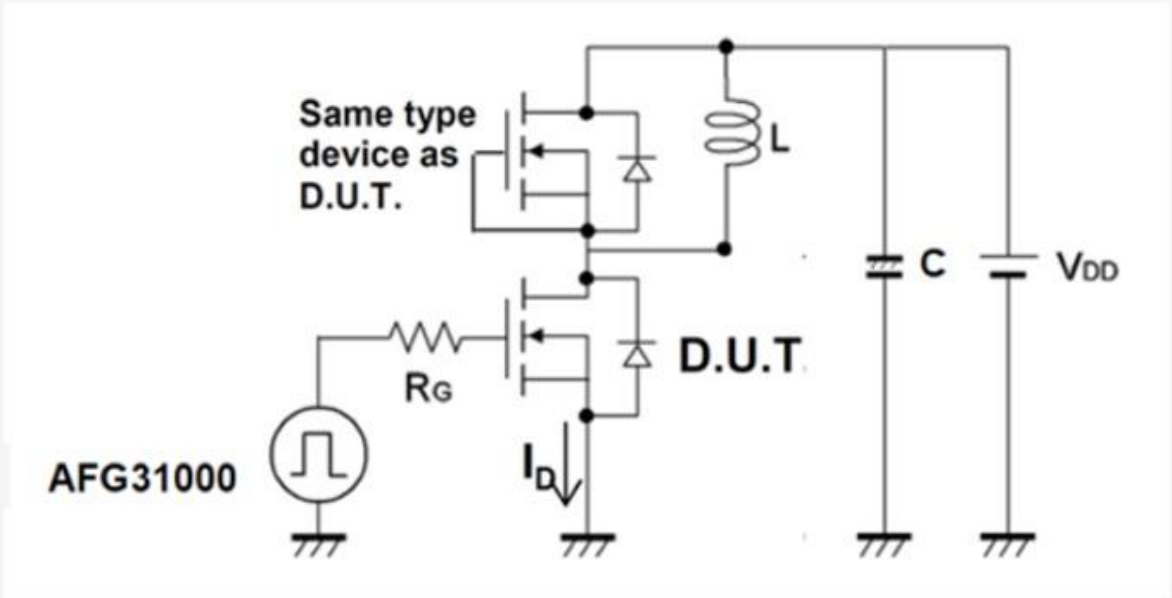
We know duty cycle = CMPA/ TBPRD *100

So, both the pulses have 50 % duty cycle.

Now, I tried to calculate the switching loss at different frequencies for the switches.

SETUP –

1- Used the lower switch q2. Kept upper switch q1 open.
2- Applied 50 kHz frequency pulse (continuous) and then increased the frequency in steps.
3- Took output voltage across the gate and source of q2 [cn2 and cn3], and output current as drain current.
4- Performed the calculation for switching loss by both hands and using the oscilloscope inbuilt maths function.



Double pulse test circuit.

The value of inductor used – 100 uH.

Value of capacitor used – 100 uF.

Vdd is kept at 10 V.

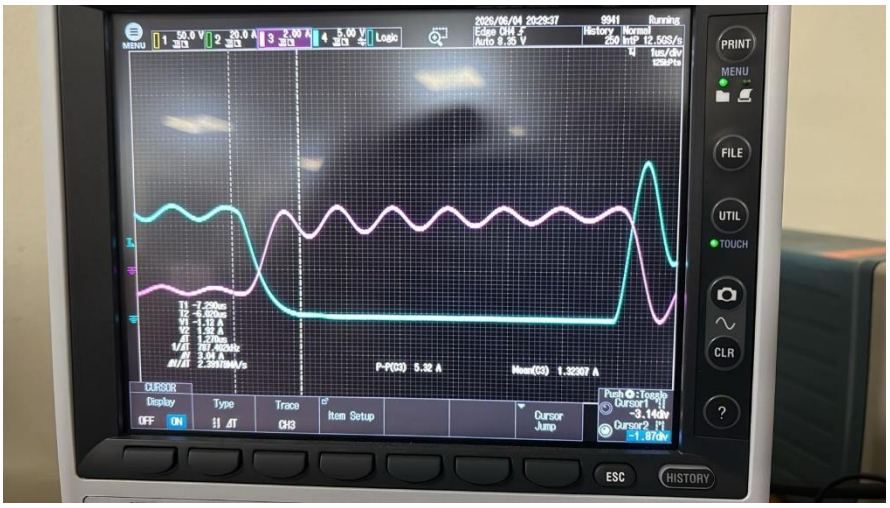
OUTPUT – VOLTAGE (blue) and CURRENT (pink)



SWITCH ON

Δ V = 11.5V

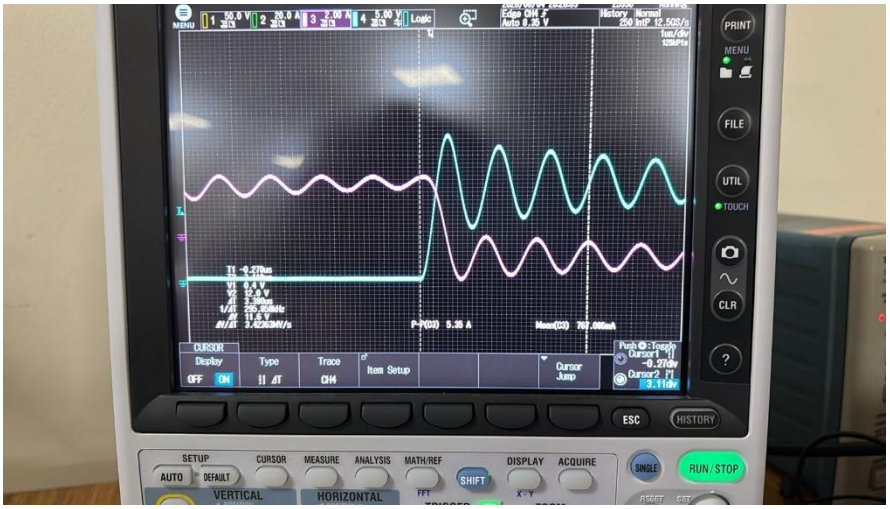
Δ T = 0.93µs



SWITCH ON

Δ I = 5.32A

Δ T = 0.93µs



SWITCH OFF

Δ V = 11.6V

Δ T = 3.380us



SWITCH OFF

Δ I = 3.21A

Δ T = 3.380us

During switching on, Turn-On time ΔT = 0.93μs is defined as time from when Vds starts falling to when Vds reaches its final low value

Turn-Off time is defined as Time from when Vds starts rising to when Vds reaches Vbus

here ΔT = 3.38μs

Manual calculations (on next page)

Switching Loss Calculation

Turn-On Energy

The turn-on energy is calculated by integrating the product of drain-source voltage and drain current over the switching transition:

$$E_{on} = \int_{t_1}^{t_2} V_{DS}(t) \cdot I_D(t)\, dt \tag{1}$$

where:

- t_1 = time at which voltage starts falling
- t_2 = time at which voltage reaches steady-state low value

Approximating the overlap as triangular, we get:

$$E_{on} = \frac{1}{2} \cdot V_{DC} \cdot I_D \cdot \Delta T \tag{2}$$

Substituting values:

$$E_{on} = \frac{1}{2} \times 11.5 \text{ V} \times 3.04 \text{ A} \times 0.93 \times 10^{-6} \text{ s} \tag{3}$$

$$\boxed{E_{on} = 16.2564 \ \mu J} \tag{4}$$

Turn-Off Energy

The turn-off energy is calculated by integrating the product of drain-source voltage and drain current over the switching transition:

$$E_{off} = \int_{t_3}^{t_4} V_{DS}(t) \cdot I_D(t)\, dt \tag{5}$$

where:

- t_3 = time at which voltage starts to rise
- t_4 = time at which voltage reaches steady-state high value

Again approximating the overlap as triangular:

$$E_{off} = \frac{1}{2} \cdot V_{DC} \cdot I_D \cdot \Delta T \tag{6}$$

Substituting values:

$$E_{off} = \frac{1}{2} \times 11.6 \text{ V} \times 3.21 \text{ A} \times 3.38 \times 10^{-6} \text{ s} \tag{7}$$

$$\boxed{E_{off} = 62.928 \ \mu J} \tag{8}$$

Total Switching Energy

$$E_{sw} = E_{on} + E_{off} = 16.2564 + 62.928 = 79.1852 \ \mu J \tag{9}$$

Switching Power Loss

At a switching frequency of $f_s = 50$ kHz:

$$P_{sw} = E_{sw} \cdot f_s \tag{10}$$

$$P_{sw} = 79.1852 \times 10^{-6} \text{ J} \times 50 \times 10^{3} \text{ Hz} \tag{11}$$

$$\boxed{P_{sw} = 3.959 \text{ W}} \tag{12}$$

Direct results from oscilloscope using the math functions

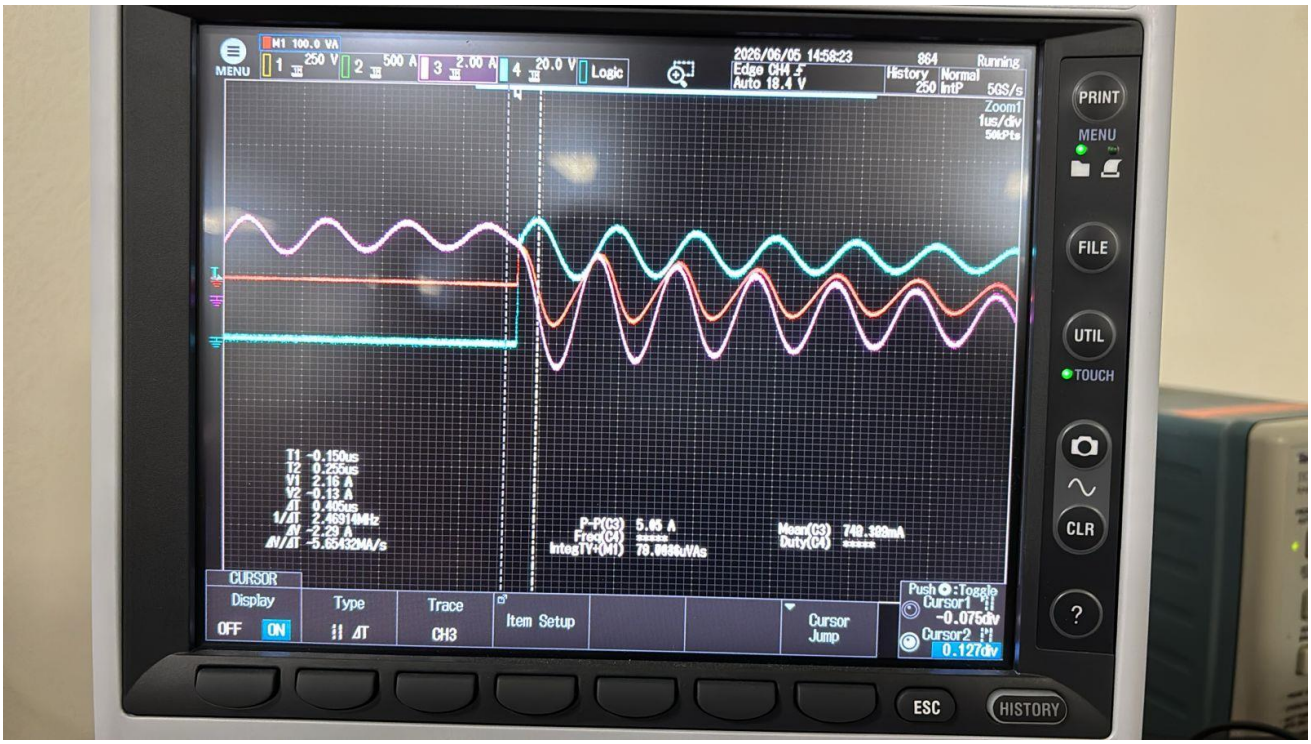
(Red line is the area under Vds*Id curve – IntegTY+)

(Blue is Vds and Pink is Id)

1 - Eon At 50 kHz = 57.0904 uVAs



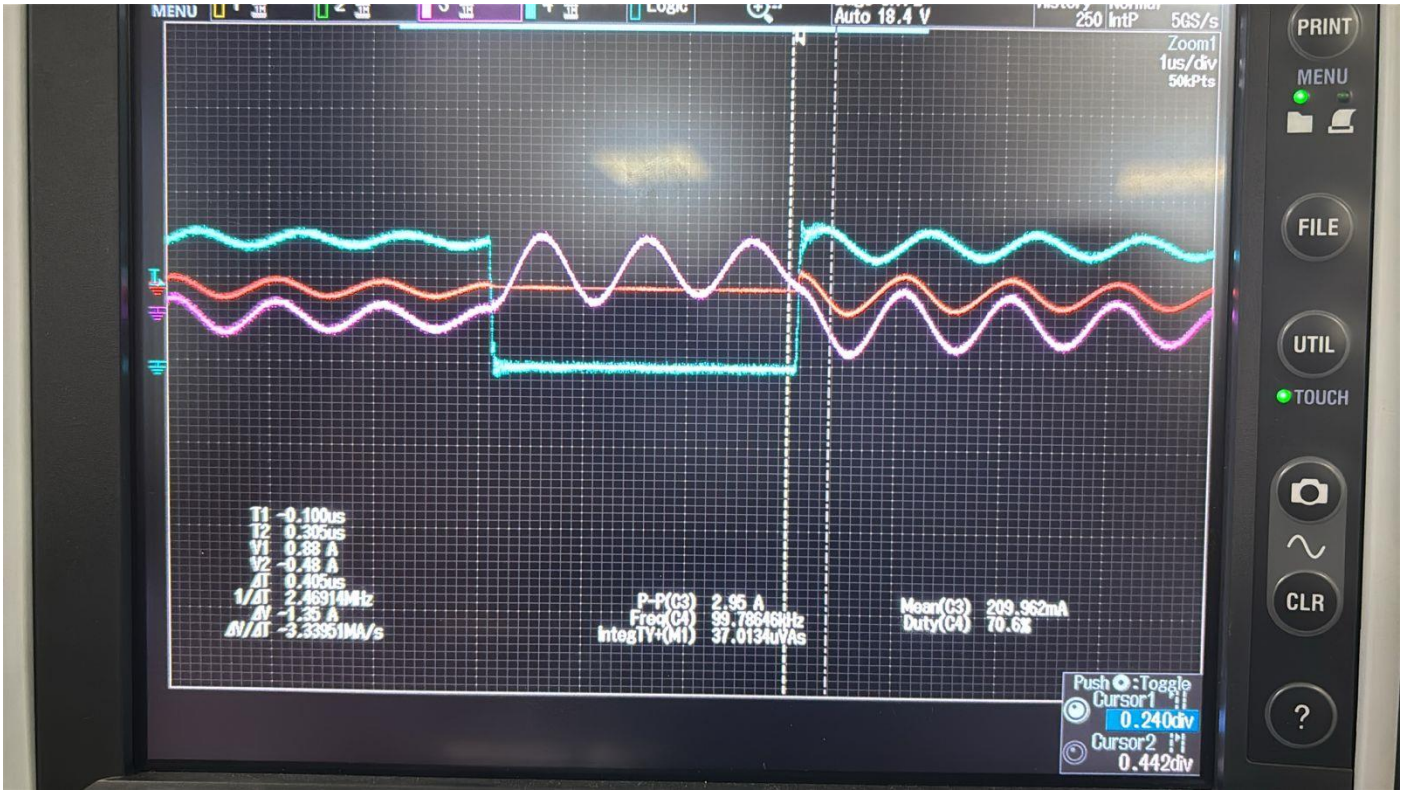
2 - Eoff at 50 kHz = 79.8686uVAs



Bus voltage 28 V and I= 0.92A during this reading.

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1 - Eoff at 100 kHz = 37.0134uVAs



2 - E on at 100 kHz= 19.5986uVAs



Bus voltage 28 V and I= 0.92A during this reading.

For calculation of switching loss.

Switching loss = (Eon+Eoff) * frequency

SiC MOSFET Switching Loss Calculation

Double Pulse Test Results

June 9, 2026

Formula

The total switching loss P_{sw} is calculated from the turn-on energy E_{on} and turn-off energy E_{off} at a given switching frequency f_{sw}:

$$P_{sw} = (E_{on} + E_{off}) \times f_{sw} \tag{1}$$

where E_{on} and E_{off} are expressed in µJ (or equivalently µV A s) and f_{sw} in kHz.

Measured Switching Energies and Calculated Losses

At 125 kHz

$$\begin{aligned} E_{on} &= 2.980\,82\,\mu J \\ E_{off} &= 2.899\,38\,\mu J \\ E_{on} + E_{off} &= 5.880\,20\,\mu J \end{aligned}$$

$$P_{sw,\,125\,kHz} = 5.880\,20\,\mu J \times 125 \times 10^3\,Hz = 0.7350\,W \tag{2}$$

At 100 kHz

$$\begin{aligned} E_{on} &= 19.5986\,\mu J \\ E_{off} &= 37.0134\,\mu J \\ E_{on} + E_{off} &= 56.6120\,\mu J \end{aligned}$$

$$P_{sw,\,100\,kHz} = 56.6120\,\mu J \times 100 \times 10^3\,Hz = 5.6612\,W \tag{3}$$

At 50 kHz

$$\begin{aligned} E_{on} &= 57.094\,\mu J \\ E_{off} &= 79.8686\,\mu J \\ E_{on} + E_{off} &= 136.9626\,\mu J \end{aligned}$$

$$P_{sw,\,50\,kHz} = 136.9626\,\mu J \times 50 \times 10^3\,Hz = 6.8481\,W \tag{4}$$

Summary Table

Table 1: Switching Loss Summary

f_{sw} (kHz)	E_{on} (µJ)	E_{off} (µJ)	E_{total} (µJ)	P_{sw} (W)
125	2.980 82	2.899 38	5.880 20	0.7350
100	19.598 60	37.013 40	56.612 00	5.6612
50	57.094 00	79.868 60	136.962 60	6.8481

Notes

- Energies measured via the Yokogawa DLM3054 oscilloscope using the IntegTY+ function on the $v_{DS} \times i_D$ math channel.
- Values in µV A s are numerically equivalent to µJ.

Problems I faced when testing for switching loss:

During the double pulse test, the measured drain-source voltage (V DS) and drain current (ID) waveforms exhibited significant oscillations following each switching transition, as observed on the Yokogawa DLM3054 oscilloscope.

These oscillations, commonly referred to as ringing, arise due to the parasitic inductance present in the PCB traces and bus connections interacting with the output capacitance (C oss) of the SIC MOSFET to form a resonant LC tank circuit.

When the device switches, the rapid change in current (dI/dt) and voltage (dV/dt) injects energy into this parasitic LC network, causing it to oscillate at its natural resonant frequency.

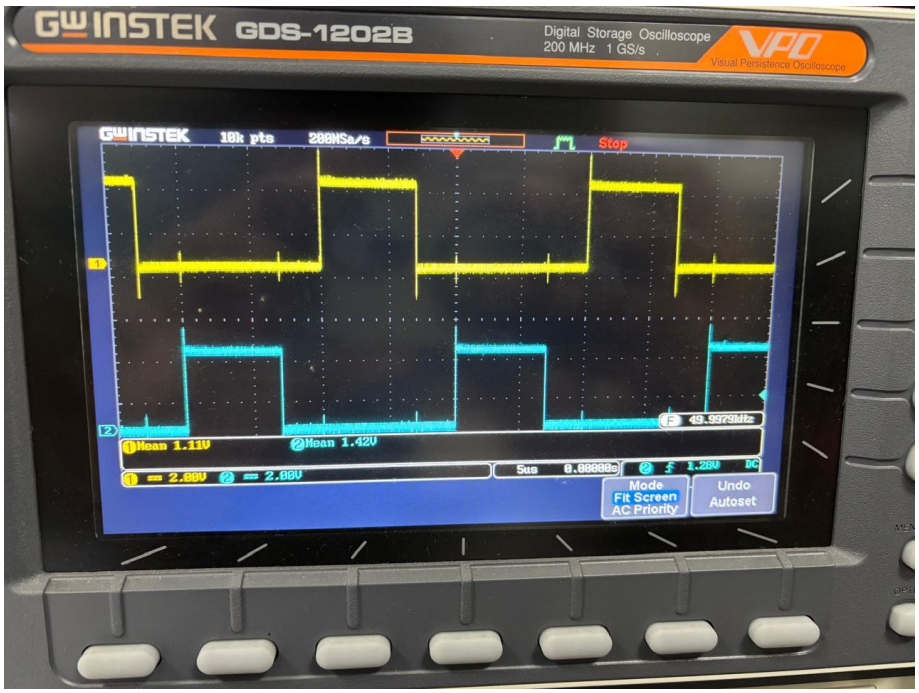
The inherently fast switching speed of SIC MOSFETs, which is significantly higher than conventional silicon devices, results in larger dV/dt and dI/dt values and consequently more pronounced ringing.

While this behaviour is expected and does not indicate a fault in the circuit, it introduces complexity in the accurate determination of switching losses, as the oscillating VDS and ID waveforms deviate from the ideal triangular overlap assumed in manual calculations. The switching energy was therefore measured directly using the IntegTY+ function on the oscilloscope, which integrates the instantaneous power waveform (VDS × ID) over the switching interval to give a more accurate result than the triangular approximation.

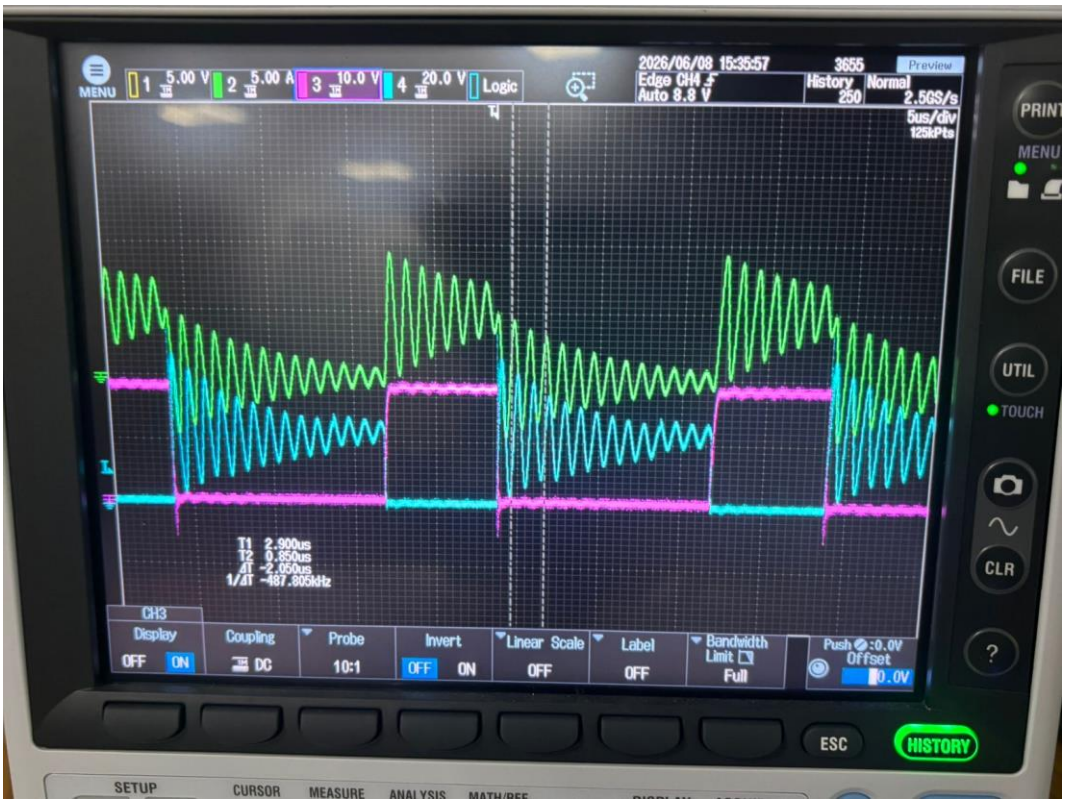
Applying the Dead Band signal to Both the Switches simultaneously.

EPWM1a – Cin+h

EPWM1b – Cin+l



INPUT dead band signal



OUTPUT taken across switch 2

blue – Vds across(Q2)

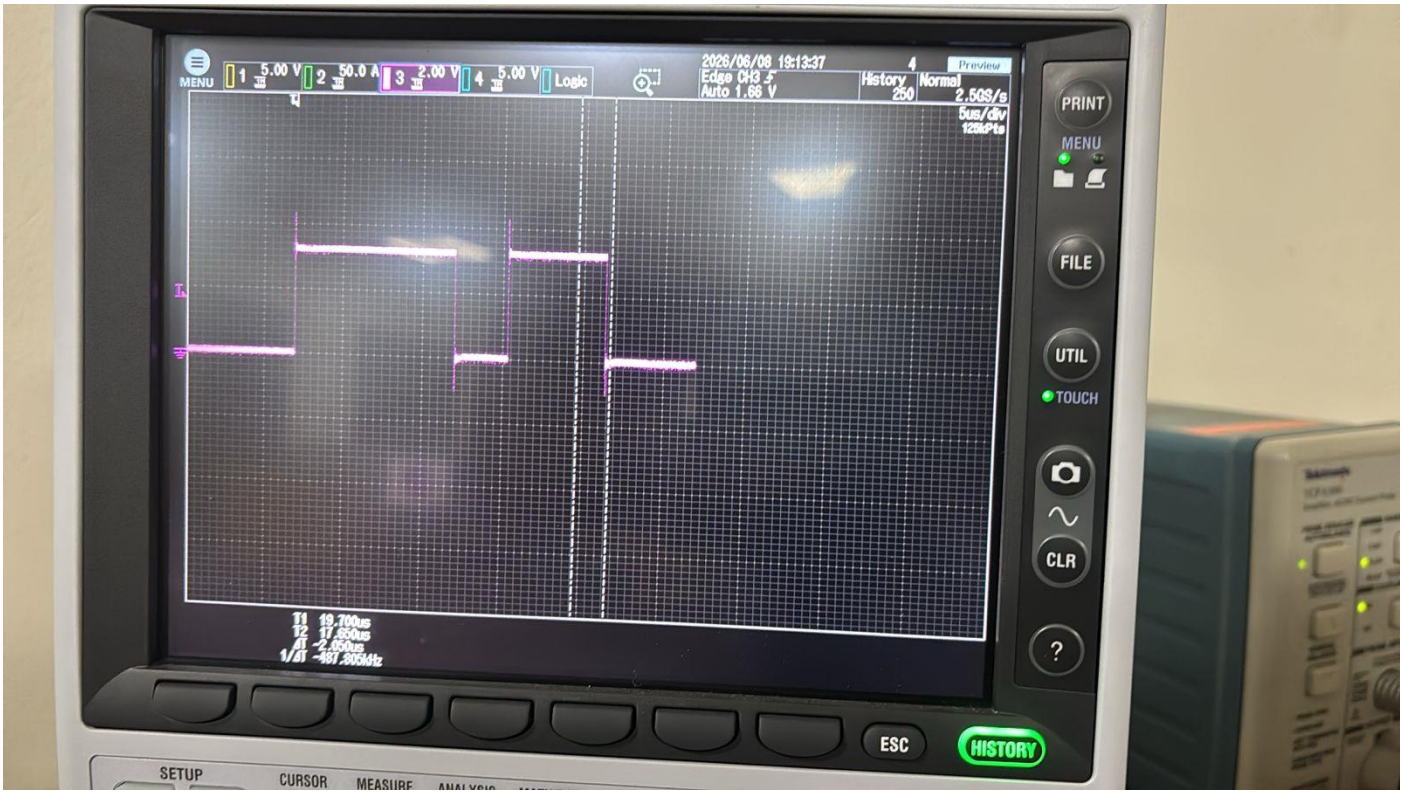
pink - gate pulse

Green- Id (Q2)

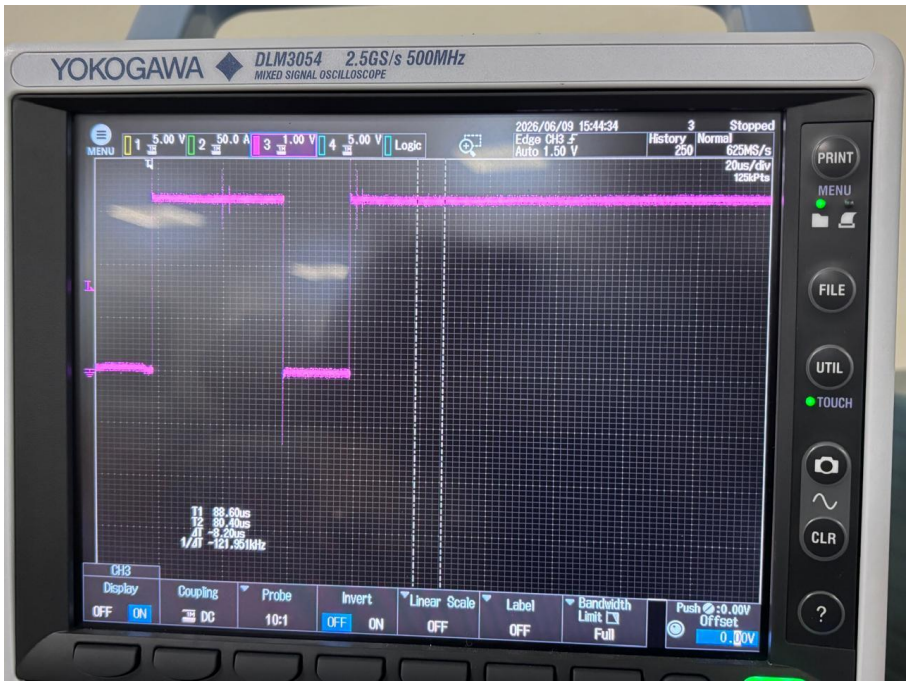
We cannot simultaneously check for both switch1 (Q1) and switch2 (Q2) because their ground will be shorted by oscilloscope. For that we require a isolated probe.

DOUBLE PULSE TEST

Finally, I captured the double pulse on oscilloscope using the single feature.

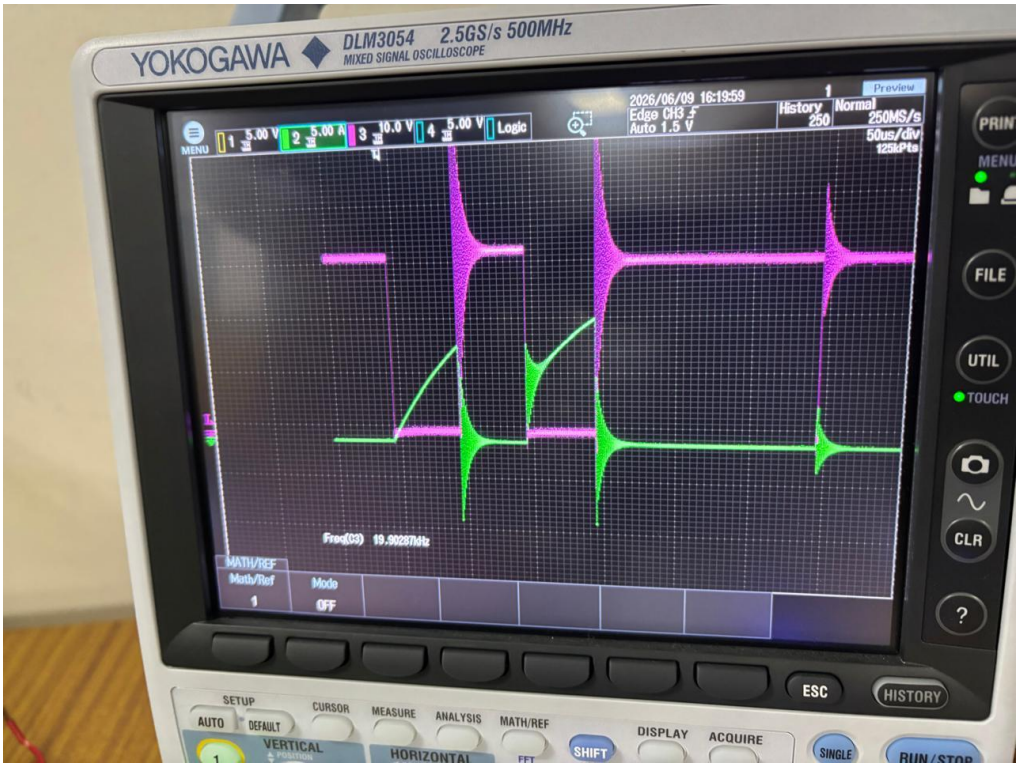


The generated signal was used to perform the Double Pulse Test. The first pulse width was 50 µs, while the second pulse width was 2.4 µs



This is the switch voltage Vds. When Vds = 0 switch is ON and current flows and when Vds is not 0 then switch is OFF and current stops.

OUTPUT OF DOUBLE PULSE TEST



Pink – Switch Voltage Vds and Green – Current (Id)



Here we used the oscilloscope's built in maths function to calculate the area under the Power (Vds*Ids) curve.

Switching loss = Eloss* fs

Where **fs is your intended converter switching frequency** — whatever frequency your final design will operate at (e.g. 50 kHz, 100 kHz, 200 kHz).

For fs = 50kHz -> Switching loss = 6.36475W (voltage probe was set to 10x)

For fs = 100kHz -> Switching loss = 12.7295W

Code used for the double pulse generation

```c
#include "F28x_Project.h"

#define TBPRD_VAL        4999
#define PULSE1_WIDTH     2500
#define PULSE2_WIDTH     120

typedef enum {
    DPT_IDLE = 0,
    DPT_PULSE1_ON,
    DPT_PULSE1_OFF,
    DPT_PULSE2_ON,
    DPT_DONE
} DPT_State;

volatile DPT_State dptState  = DPT_IDLE;
volatile Uint32    tickCount = 0;

void InitEPWM1_DPT(void);
void GPIO_Init(void);
interrupt void epwm1_isr(void);

void main(void)
{
    InitSysCtrl();

    DINT;
    InitPieCtrl();
    IER = 0x0000;
    IFR = 0x0000;
    InitPieVectTable();

    GPIO_Init();
    InitEPWM1_DPT();

    EALLOW;
    PieVectTable.EPWM1_INT = &epwm1_isr;
    EDIS;

    IER |= M_INT3;
    PieCtrlRegs.PIEIER3.bit.INTx1 = 1;

    EINT;
    ERTM;

    EALLOW;
    EPwm1Regs.CMPA.bit.CMPA = PULSE1_WIDTH;
    EDIS;
```

```
48
49      dptState  = DPT_PULSE1_ON;
50      tickCount = 0;
51
52      EPwm1Regs.TBCTL.bit.CTRMODE = TB_COUNT_UP;
53
54      while(1)
55      {
56          __asm("␣NOP");
57      }
58  }
59
60  interrupt void epwm1_isr(void)
61  {
62      tickCount++;
63
64      switch(dptState)
65      {
66          case DPT_PULSE1_ON:
67              EALLOW;
68              EPwm1Regs.CMPA.bit.CMPA = 0;
69              EDIS;
70              dptState  = DPT_PULSE1_OFF;
71              tickCount = 0;
72              break;
73
74          case DPT_PULSE1_OFF:
75              if(tickCount >= 1)
76              {
77                  EALLOW;
78                  EPwm1Regs.CMPA.bit.CMPA = PULSE2_WIDTH;
79                  EDIS;
80                  dptState  = DPT_PULSE2_ON;
81                  tickCount = 0;
82              }
83              break;
84
85          case DPT_PULSE2_ON:
86              EALLOW;
87              EPwm1Regs.CMPA.bit.CMPA      = 0;
88              EPwm1Regs.AQCTLA.bit.CAU     = AQ_NO_ACTION;
89              EPwm1Regs.AQCTLA.bit.ZRO     = AQ_CLEAR;
90              EPwm1Regs.AQSFRC.bit.ACTSFA = AQ_CLEAR;
91              EPwm1Regs.AQSFRC.bit.OTSFA  = 1;
92              EPwm1Regs.TBCTL.bit.CTRMODE = TB_FREEZE;
93              EDIS;
94              dptState = DPT_DONE;
```

```
            break;

        case DPT_DONE:
        default:
            break;
    }

    EPwm1Regs.ETCLR.bit.INT = 1;
    PieCtrlRegs.PIEACK.all  = PIEACK_GROUP3;
}

void InitEPWM1_DPT(void)
{
    EALLOW;
    CpuSysRegs.PCLKCR2.bit.EPWM1            = 1;
    ClkCfgRegs.PERCLKDIVSEL.bit.EPWMCLKDIV = 1;
    CpuSysRegs.PCLKCR0.bit.TBCLKSYNC        = 0;
    EDIS;

    EPwm1Regs.TBCTL.bit.CTRMODE   = TB_FREEZE;
    EPwm1Regs.TBCTL.bit.PHSEN     = TB_DISABLE;
    EPwm1Regs.TBCTL.bit.HSPCLKDIV = TB_DIV1;
    EPwm1Regs.TBCTL.bit.CLKDIV    = TB_DIV1;
    EPwm1Regs.TBCTL.bit.SYNCOSEL  = TB_CTR_ZERO;
    EPwm1Regs.TBPRD               = TBPRD_VAL;
    EPwm1Regs.TBCTR               = 0;

    EPwm1Regs.CMPA.bit.CMPA        = 0;
    EPwm1Regs.CMPCTL.bit.SHDWAMODE = CC_SHADOW;
    EPwm1Regs.CMPCTL.bit.LOADAMODE = CC_CTR_ZERO;

    EPwm1Regs.AQCTLA.bit.ZRO = AQ_SET;
    EPwm1Regs.AQCTLA.bit.CAU = AQ_CLEAR;

    EPwm1Regs.ETSEL.bit.INTSEL = ET_CTR_ZERO;
    EPwm1Regs.ETSEL.bit.INTEN  = 1;
    EPwm1Regs.ETPS.bit.INTPRD  = ET_1ST;

    EALLOW;
    CpuSysRegs.PCLKCR0.bit.TBCLKSYNC = 1;
    EDIS;
}

void GPIO_Init(void)
{
    EALLOW;
    GpioCtrlRegs.GPAMUX1.bit.GPIO0 = 1;
    GpioCtrlRegs.GPADIR.bit.GPIO0  = 1;
    EDIS;
}
```

Listing 1: Standard Double Pulse Test – TMS320F28379D

Conclusion

This internship provided valuable hands-on experience in the design and characterization of a SIC MOSFET gate driver system, covering the complete workflow from schematic design and PCB development to hardware testing and switching loss measurement. The work helped bridge the gap between theoretical concepts learned in coursework and their practical implementation in power electronics.

Designing the PCB in Altium Designer highlighted the importance of component placement, trace routing, and design rule checks (DRC), especially for high-frequency switching applications. Although exact SPICE models for some components were unavailable, LTspice simulations were useful in understanding the expected switching behavior of the gate driver before hardware implementation.

Working with the TMS320F28379D C2000 microcontroller improved my understanding of control systems. I gained practical experience with ePWM configuration, including clock settings, frequency generation using TBPRD, duty cycle control through CMPA, and deadband insertion for complementary gate signals.

The Double Pulse Test was used to study the switching performance of the SCT35N65 SIC MOSFET. The measured switching energies were 57.09 μJ during turn-on and 79.87 μJ during turn-off at 50 kHz operation. The results showed the typical difference between turn-on and turn-off losses observed in SIC devices. Ringing observed after switching also demonstrated the effect of parasitic inductance and capacitance, emphasizing the importance of careful PCB layout in high-frequency power converters.

Overall, this internship provided exposure to the complete development cycle of a power electronics system—from design and simulation to microcontroller programming and experimental testing. The experience strengthened my technical knowledge and practical skills, and it has increased my interest in further research and applications involving wide bandgap devices, motor drives, and high-efficiency power converters.
